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Invited Review

A survey of berth allocation and quay crane scheduling problems in container terminals

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ABSTRACT

Due to the variety of technical equipments and terminal layouts, research has produced a multitude of optimization models for seaside operations planning in container terminals. To provide a support in modeling problem characteristics and in suggesting applicable algorithms this paper reviews the relevant literature. For this purpose new classification schemes for berth allocation problems and quay crane scheduling problems are developed. Particular focus is put on integrated solution approaches which receive increasing importance for the terminal management.

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1. Introduction

In recent years, OR methods have received considerable importance for the operations management in container terminals (CTs). Comprehensive overviews on applications and optimization models in this field are given by Meersmans and Dekker (2001), Vis and de Koster (2003), Steenken et al. (2004), Vacca et al. (2007), and Stahlbock and Voß (2008). A significant amount of papers dealing with the application of OR methods addresses the planning of the seaside transshipment operations. Fig. 1 shows important relations of the strategic planning and the operations planning at the seaside area, the yard, and the landside area.

One issue of seaside operations planning is the assignment of quay space and service time to vessels that have to be unloaded and loaded at a terminal. This problem is commonly referred to as the berth allocation problem (BAP). The transshipment of containers between a vessel and the quay is generally performed by specialized cranes, which are mounted on rail tracks alongside the quay. The assignment of these quay cranes (QCs) to vessels and the determination of work plans for the cranes addresses two further problems, namely the quay crane assignment problem (QCAP) and the quay crane scheduling problem (QCSP). Solutions to these problems must respect the berth layout and the used equipment, whereas they impact the yard operations and the workforce planning, see Fig. 1.

Due to the variety of technical equipments and terminal layouts, research has produced a multitude of optimization models for the BAP, the QCAP, and the QCSP. Moreover, a trend towards an integrated solution of these problems is observed in the recent literature. The large number of available models and proposed solution methods prevents an easy choice of a suitable approach in a specific situation. To provide a support in modeling problem characteristics and in suggesting applicable algorithms this paper develops classification schemes for BAPs, QCSPs, and integrated approaches.

The paper is organized as follows. In Section 2 the focused operational planning problems are described in detail against the background of different terminal properties and objectives. A literature survey of BAP and QCAP formulations is presented in Section 3 which is derived from a new classification scheme for these problems. Accordingly, a classification scheme and a literature survey are presented for QCSP formulations in Section 4. Since future advances in the field are expected from integrated solution approaches, Section 5 provides a literature review of the state-of-the-art integration concepts. The paper is summarized in Section 6.

2. Planning of seaside operations

2.1. Berth allocation problem

In the BAP we are given the berth layout of a CT together with a set of vessels that have to be served within the planning horizon. For each vessel additional data like the vessel's length including clearance, its draft, the expected time of arrival, and the projected

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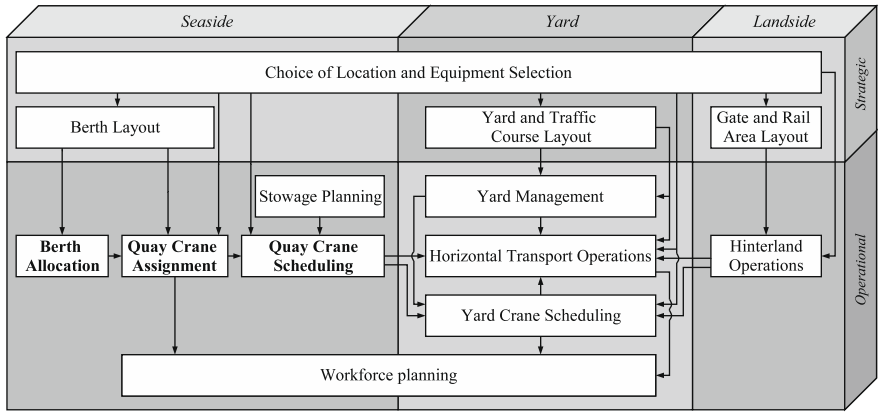

Fig. 1. Planning problems in container terminals.

handling time can be given. All vessels must be moored within the boundaries of the quay. They are not allowed to occupy the same quay space at a time. The problem is to assign a berthing position and a berthing time to each vessel, such that a given objective function is optimized. An example for the graphical representation of a berth plan with five vessels is shown in Fig. 2a. Berth planning has been shown to be an $\mathcal{NP}$-hard problem by relating it to the set partitioning problem (Lim, 1998), the single machine scheduling problem with release dates (Hansen and Oğuz, 2003), and the two dimensional cutting stock problem (Imai et al., 2005).

There may be further constraints involved in berth allocation, which leads to a multitude of BAP formulations. Spatial constraints restrict the feasible berthing positions of vessels according to a preset partitioning of the quay into berths. According to Imai et al. (2005) the following cases are distinguished:

(a) Discrete layout: The quay is partitioned into a number of sections, called berths. Only one vessel can be served at each single berth at a time. The partitioning can either follow the construction of the quay (Fig. 3a) or is organizationally prescribed to ease the planning problem (Fig. 3b).
(b) Continuous layout: There is no partitioning of the quay, i.e. vessels can berth at arbitrary positions within the boundaries of the quay (Fig. 3c). For a continuous layout, berth planning is more complicated than for a discrete layout at the advantage of better utilizing quay space.
(c) Hybrid layout: Like in the discrete case, the quay is partitioned into berths, but large vessels may occupy more than one berth (Fig. 3d) while small vessels may share a berth (Fig. 3e). An indented berth results if two opposing berths exist, which can be used to serve a large vessel from both sides (Fig. 3f).

In case of draft restrictions further spatial constraints must ensure that vessels are berthed at positions of sufficient water depth. To avoid enlarging the handling time, container vessels usually stay at the assigned berthing position during the entire service. In contrast, vessels are allowed to be repositioned in naval ports and general cargo terminals, see e.g. Brown et al. (1994, 1997) and Lee and Chen (2008).

Temporal constraints can restrict the berthing times and the departure times of vessels. According to Imai et al. (2001) the following cases are distinguished:

(a) Static arrival: There are no arrival times given for the vessels or arrival times impose merely a soft constraint on the berthing times. In the former case it is assumed that vessels already wait at the port and can berth immediately. In the latter case it is assumed that a vessel can be speeded up at a certain cost in order to meet a berthing time earlier than the expected arrival time.
(b) Dynamic arrival: Fixed arrival times are given for the vessels, hence, vessels cannot berth before the expected arrival time.

In order to keep liner schedules, latest departure times of the vessels can be prescribed additionally. In some papers due dates are expressed by a maximum waiting plus handling time for a vessel. In the dynamic case, the entire service of a vessel must be executed within the resulting time window.

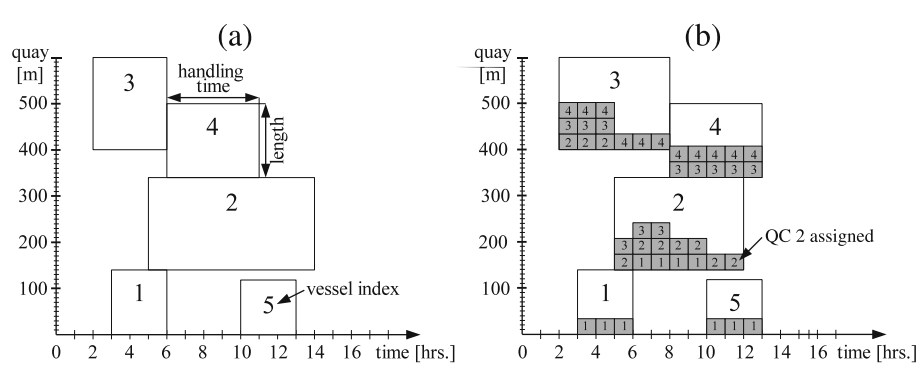

Fig. 2. Space-time representation of a berth plan (a), assignment of cranes to vessels (b).

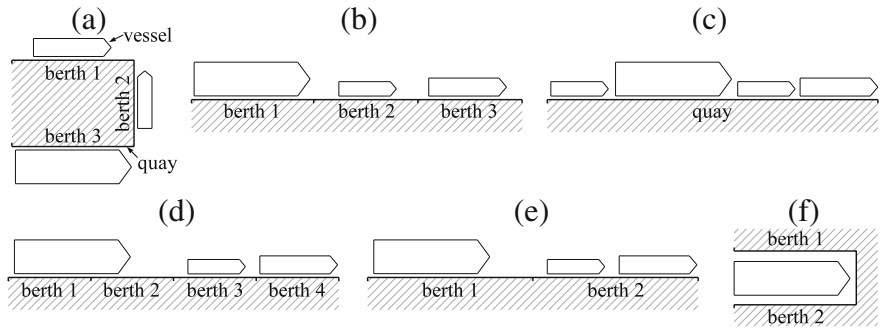

Fig. 3. Berth and quay relationship.

Vessel handling times are assumed deterministic in the vast majority of published BAP models. Still, literature deals with vessel handling times in different ways:

(a) They are known in advance and considered unchangeable, i.e. they are fixed.
(b) They depend on the vessels' berthing positions.
(c) They depend on the number of cranes serving the vessels.
(d) They depend on the work schedules of the assigned cranes.
(e) They obey to combinations of (b), (c), and (d).

The general goal of berth planning is to provide fast and reliable services of vessels. This is reflected in the literature by various objective functions. Models to minimize the sum of the waiting and handling times of vessels (i.e. the port stay times) clearly prevail. Further objectives are, for example, the minimization of the workload of terminal resources and the minimization of the number of vessels rejected to be served at a terminal. The performance of a berth plan is often measured in terms of costs which allows to combine different goals in an overall cost function.

2.2. Quay crane assignment problem

In the QCAP we are given a feasible berth plan and a set of identical QCs, which are available for service. For each vessel included in the berth plan, the volume of containers to be loaded and unloaded is known as well as the maximum number of cranes allowed to serve it simultaneously. The cranes are supposed to be lined up alongside the quay. They can be moved to every vessel but they are not able to pass each other. The problem is to assign cranes to vessels such that all required transshipments of containers can be fulfilled, see Fig. 2b. The distribution of cranes to vessels is sometimes referred to as the crane split, cf. Steenken et al. (2004). QCAP and BAP are basically interrelated, because solving the QCAP can have a strong impact on the vessels' handling times. Only in case of a discrete berth layout, where each berth holds a set of dedicated cranes, an explicit assignment of cranes to vessels is not necessary.

A quay crane assignment problem can come along with specific requirements:

(a) Either a number of cranes or a set of specific cranes is assigned to a vessel.
(b) The number of cranes assigned to a vessel is unchangeable throughout the service process, which is referred to as a time-invariant assignment (shown for Vessels 1, 4, and 5 in Fig. 2b). In contrast, a variable-in-time assignment of cranes (Vessels 2 and 3 in Fig. 2b) is enabled in some approaches.
(c) A minimum number of cranes serving the vessel throughout the whole handling time can be prescribed. This number is usually part of the contract between the vessel operator and the CT operator.

Crane capacity as well as a transshipment volume of containers are usually measured in terms of QC-hours. At least for transshipment volumes, this is not a precise measure because interference of cranes can prevent an effective use of the assigned QC-hours. Therefore, a decreasing marginal crane productivity is modeled in some approaches. Typical objectives of the QCAP aim at minimizing crane productivity losses by reducing the number of crane setups at vessels and the crane travel times.

In practice, the QCAP is not found a difficult problem if solved by rules of thumb. Therefore, the problem has hardly received attention by its own in academic research. Due to the profound impact on vessels' handling times, however, crane assignment decisions are involved in some advanced berth planning models.

2.3. Quay crane scheduling problem

In the QCSP we consider a set of tasks, representing transshipment operations for a vessel, and a set of assigned QCs. Precedence relations among tasks can be given to ensure that unloading precedes loading and to represent the stacking of containers as defined by a stowage plan. Every task must be processed (usually without preemption) once by a QC while a QC can process at most one task at a time. A solution to the problem, called a QC schedule, defines a starting time for every task on a crane. Usually, the minimization of the makespan of the QC schedule is pursued because it represents the handling time of the considered vessel. The problem described thus far corresponds to a minimum makespan scheduling problem with parallel identical machines and precedence constraints. This problem is known to be $\mathcal{NP}$-hard in the strong sense, provided that more than two machines (cranes), non-preemption or non-uniform processing times are given (Pinedo, 2002).

To avoid crossing of cranes, the QCSP requires a spatial constraint, which is not involved in machine scheduling problems. As a further spatial constraint, sophisticated QCSP models also comprise the compliance with safety margins between adjacent cranes. Additional attributes for tasks and cranes lead in fact to a variety of different models for QC scheduling.

Tasks to be scheduled on a QC describe the granularity in which the workload of a vessel is considered in a QCSP model. Tasks can be defined on the basis of bay areas or single bays (Fig. 4a), or on the basis of container stacks, container groups, or individual containers (Fig. 4b):

(a) Bay areas: A task consists of all loading and unloading operations of containers within a certain area of the bays.
(b) Bays: A task consists of all loading and unloading operations in a bay.
(c) Stacks: A task consists of the transshipment of all containers in a stack.

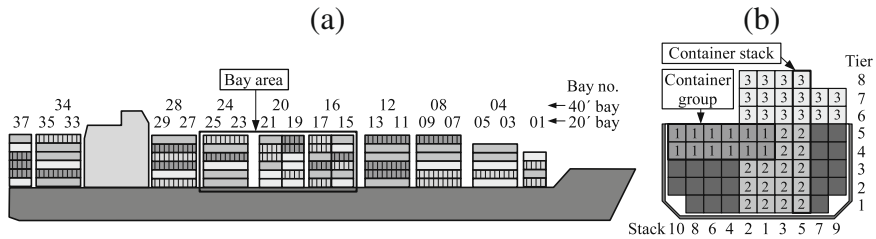

Fig. 4. Storage location structure of a vessel (a) and a cross-sectional view of a bay (b).

(d) Groups: A task refers to a group of containers that are stored in adjacent slots of a bay. Grouped containers usually have a common destination, or the like.
(e) Containers: A task consists of the loading or unloading of a single container.

The idea of dividing the workload of a vessel into bay areas is to serve each bay area exclusively by one QC. If the bay areas are non-overlapping, crane interference is completely avoided. However, a sufficient balance of the workload distribution among the cranes is often not possible. In case (b), a better workload distribution might be achievable, but solving the problem becomes much more complicated due to the required non-crossing constraint. As the number of tasks is bounded by the size of the vessel, the problem complexity is still moderate. Reducing the granularity further on allows to improve crane schedules at the expense of increasing the problem complexity. With hundreds to thousands of tasks, as observed for a large vessel in case (e), the corresponding QCSP might become intractable.

Next to task attributes, also crane attributes appear in QCSP models to specify the crane operations in more detail:

(a) Ready times: An individual ready time is used to designate the earliest possible operation of a crane.
(b) Time windows: For each crane one or more time windows can be used to specify time spans where the crane is available to serve the considered vessel. These time windows are often an outcome of variable-in-time crane assignments.
(c) Positions: Initial and final positions are prescribed for a crane.
(d) Travel times: The speed of the crane movement is given in terms of the time required to travel between bays.

2.4. Problem integration approaches

Basically, berth allocation, QC assignment, and QC scheduling decisions can be made in a sequential fashion as shown in Fig. 5. This way the overall problem complexity of seaside operations planning is broken down into a series of decisions. Nevertheless, existing interrelations between the planning levels are almost completely ignored by sequential planning. Often, this leads to plans of poor overall quality. If, for example, the vessel handling times are assumed too tight in the BAP, the QCSP might get infeasible. On the other hand, if the handling times are too loose, crane schedules will show a lot of slack time that cannot be used productively anymore.

A remedy is promised by considering seaside operations planning within a monolithic optimization problem. In practical environments, however, such models are much too huge to be solvable at all. Therefore, recent research focuses on the design of appropriate problem integration approaches. According to Geoffrion (1999) integration of problems can be done either by a *deep integration* or by a *functional integration*. Similar concepts are proposed by Muhanna and Pick (1988) and Dolk and Kottemann (1993) in different terms.

Deep integration merges two subproblems into a partial monolithic problem formulation which makes the propagation of instructions and the explicit consideration of relations between these problems obsolete. Deep integration can be used for instance to merge a BAP and a QCAP into a combined problem of berth allocation and crane assignment.

A functional integration is realized by a computational agenda that defines a sequence for solving subproblems and data exchanges between the base-level and the top-level. Two problems are functionally integrated either by incorporating a feedback loop structure or a preprocessing phase. In a feedback loop structure (Fig. 6a), the top-level problem instructs the base-level problem and the base-level reaction is used in the top-level to revise instructions. This loop terminates once a steady state is reached. For example, in a loop where the QCAP and the QCSP are involved, the QCAP determines crane assignments for vessels served simultaneously. For each vessel an individual QCSP instance results, which is solved at the base-level. At the top-level the returned crane schedules are used to check whether sufficient crane capacity is assigned to the vessels. Functional integration by preprocess-

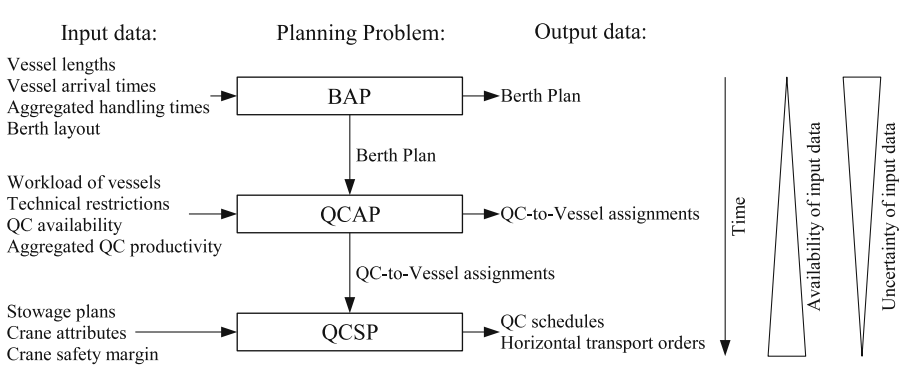

Fig. 5. Sequential planning of seaside operations.

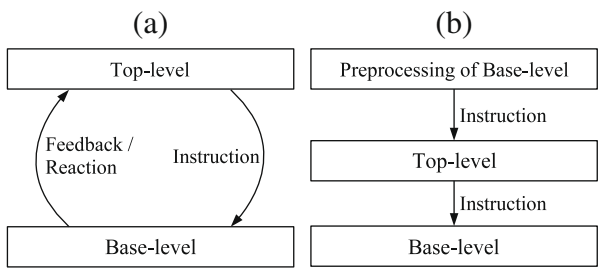

Fig. 6. Functional integration by a feedback loop (a) and by a preprocessing (b).

ing (Fig. 6b) starts solving the base-level problem in order to generate more detailed input data for the top-level problem. Afterward, both problems are solved sequentially in the common way. For example, given a promising crane assignment, a set of corresponding QCSP instances is solved. The makespans of crane schedules, which are generated in the preprocessing, are taken as vessel handling times in the berth planning phase.

Different designs for integrated seaside operations planning are possible involving the above described integration concepts and the BAP, QCAP, and QCSP as subproblems.

3. Berth allocation and quay crane assignment

3.1. Classification scheme

To show similarities and differences in the existing models for berth allocation, a classification scheme is developed in the following. Studies that concentrate on quay crane assignment either presuppose a particular type of BAP or integrate quay crane assignment decisions in the berth planning process. For this reason, QCAP approaches are captured by the classification scheme as well. Problems are classified according to four attributes. The *spatial attribute* concerns the berth layout and water depth restrictions. The *temporal attribute* describes the temporal constraints for the service process of vessels. The *handling time attribute* determines the way vessel handling times are considered in the problem. The fourth attribute defines a *performance measure* for evaluating possible solutions to a problem. Each attribute can take different values. They are listed in Table 1.

Spatial, temporal, and handling time attributes have been described in Section 2.1. The performance measures listed in Table 1 reflect different service quality criteria. Minimizing the waiting time or the handling time of a vessel aims at providing a competitive service to vessel operators. If both objectives are pursued (i.e. *wait* and *hand* are set), the port stay time of vessels is minimized. Minimizing the completion times of vessels (*compl*) aims at earliest possible departures. In the presence of soft arrival times or soft due dates either the speedup of vessels (*speed*) or the tardiness of vessels (*tard*) has to be minimized. The *order* measure strives at a reduction of the deviation between the arrival order of vessels and the planned service order. It is assessed by the number of vessels not served in First-come First-served manner. If it is foreseeable that a vessel cannot be served within the desired time window, it may be rejected at the terminal (and possibly reassigned to another terminal of the port). Hence, the minimization of vessel rejections (*rej*) is considered as a goal in some models. If labor or other resources are scarce at a terminal, the resource utilization (*res*) is optimized. The minimization of deviations between chosen berthing positions and desired positions (*pos*) aims at reducing the travel distances for the horizontal transport vehicles. If none of the above performance measures is used in a BAP formulation, the value *misc* (miscellaneous) is set in the classification.

Table 1
A classification scheme for BAP formulations.

Value	Description
1. Spatial attribute	
disc	The quay is partitioned in *disc*rete berths
cont	The quay is assumed to be a *cont*inuous line
hybr	The *hybr*id quay mixes up properties of discrete and continuous berths
draft	Vessels with a draft exceeding a minimum water depth cannot be berthed arbitrarily
2. Temporal attribute	
stat	In *stat*ic problems there are no restrictions on the berthing times
dyn	In *dyn*amic problems arrival times restrict the earliest berthing times
due	*Due* dates restrict the latest allowed departure times of vessels
3. Handling time attribute	
fix	The handling time of a vessel is considered *fix*ed
pos	The handling time of a vessel depends on its berthing *pos*ition
QCAP	The handling time of a vessel depends on the assignment of QCs
QCSP	The handling time of a vessel depends on a QC operation schedule
4. Performance measure	
wait	*Wait*ing time of a vessel
hand	*Hand*ling time of a vessel
compl	*Compl*etion time of a vessel
speed	*Speed*up of a vessel to reach the terminal before the expected arrival time
tard	*Tard*iness of a vessel against the given due date
order	Deviation between the arrival *order* of vessels and the service order
rej	*Rej*ection of a vessel
res	*Res*ource utilization effected by the service of a vessel
pos	Berthing of a vessel apart from its desired berthing *pos*ition
misc	*Misc*ellaneous

The above listed measures address criteria to be minimized. Either the minimization of the total measure for all vessels or the minimization of the measure for the worst performing vessel can serve as an objective function in a BAP. A total measure is denoted in the classification scheme by a $\sum()$ function and a worst performing measure, i.e. a min–max objective, is denoted by a $max()$ function. Vessel specific weights are indicated by the denotation w. Moreover, if weights appear with an index, i.e. w_1 to w_4, they address weights of combined performance measures.

Using the introduced classification scheme, a certain type of BAP is described by a selection of values for each of the four attributes. As an example, consider a problem where the quay is partitioned into discrete berths serving the vessels exclusively (*disc*). The arrival times restrict the earliest berthing of vessels (*dyn*) and handling times are known and fixed (*fix*). The objective is to minimize the total cost arising for tardiness of vessels (*tard*) and for berthing vessels apart from desired berthing positions (*pos*). Different cost rates (w_1, w_2) apply to these performance measures. According to the proposed scheme, this problem is classified by $disc|dyn|fix|\sum(w_1 tard + w_2 pos)$. In case that the maximum tardiness of vessels has to be minimized, the problem is classified by $disc|dyn|fix|max(tard)$.

3.2. Problem classification

Table 2 gives a comprehensive survey of berth allocation and quay crane assignment formulations from the literature. Some authors outline approaches more or less informally while others provide precise optimization models. If a unique classification of a paper is not possible according to the given information, the best fit of classifying attributes is taken. The classification exclusively covers research dealing with the operational decisions regarding the BAP and QCAP. Not covered are studies employing analytical models, simulation, and queuing theory as is used for the evaluation of investment decisions and berthing policies, and for the determination of terminal throughput and system dynamics of CTs, see Edmond and Maggs (1978), Schonfeld and Frank (1984),

Table 2
Overview of BAP formulations.

Problem classification	Reference
$disc\|stat\|pos\|\sum(wait+hand)$	Imai et al. (2001) Hansen and Oguz (2003)
$disc\|stat\|pos\|\sum(wait+hand+w_1order)$	Imai et al. (1997)
$disc\|stat,due\|pos\|\sum wrej$	Imai et al. (2008b)
$disc\|stat\|pos,QCSP\|\sum(wait+hand)$	Lee et al. (2006)
$disc\|dyn\|pos\|\sum(wait+hand)$	Imai et al. (2001) Hansen and Oğuz (2003) Monaco and Sammarra (2007)
$disc\|dyn\|pos\|\sum w(wait+hand)$	Imai et al. (2003) Theofanis et al. (2007a)
$disc\|dyn\|pos\|\sum(w_1wait+w_2tard+w_3pos)$	Hansen et al. (2008)
$disc\|dyn\|pos\|\sum wtard$	Golias et al. (2006)
$disc\|dyn\|pos\|\sum wtard,\sum(wait+hand)$	Imai et al. (2007b)
$disc\|dyn\|pos\|misc$	Golias et al. (2007)
$disc\|dyn,due\|pos\|\sum w(wait+hand)$	Cordeau et al. (2005) Mauri et al. (2008)
$disc\|dyn,due\|pos\|\sum wrej$	Imai et al. (2008b)
$disc,draft\|dyn\|pos\|\sum(wait+hand)$	Han et al. (2006)
$disc,draft\|dyn,due\|pos\|\sum wait$	Zhou et al. (2006)
$disc\|dyn\|pos,QCAP\|\sum(wait+hand)$	Imai et al. (2008a)
$disc\|dyn\|QCAP\|\sum(wait+hand+tard)$	Liang et al. (2008)
$disc\|dyn,due\|QCAP\|-\sum(w_1res-w_2pos)$	Giallombardo et al. (2008)
$cont\|stat\|fix\|max(compl)$	Li et al. (1998)
$cont\|stat\|fix\|\sum wcompl$	Guan et al. (2002)
$cont\|stat\|QCAP\|\sum(w_1wait+w_2speed+w_3tard+w_4pos)$	Park and Kim (2003) Rashidi (2006)
$cont\|stat\|QCAP\|max(compl)$	Oğuz et al. (2004)
$cont\|dyn\|fix\|\sum w(wait+hand)$	Guan and Cheung (2004)
$cont\|dyn\|fix\|\sum(w_1wait+w_2pos+w_3rej)$	Wang and Lim (2007)
$cont\|dyn\|fix\|\sum(w_1tard+w_2pos)$	Moon (2000) Park and Kim (2002) Kim and Moon (2003) Briano et al. (2005)
$cont\|dyn\|fix\|max(res)$	Lim (1998)
$cont,draft\|dyn\|fix\|max(res)$	Lim (1999) Tong et al. (1999) Goh and Lim (2000)
$cont,draft\|dyn\|pos\|\sum(wait+hand)$	Chang et al. (2008)
$cont\|dyn\|pos\|\sum(wait+hand)$	Imai et al. (2005)
$cont\|dyn\|pos,QCAP\|\sum(w_1speed+w_2tard+w_3res)$	Meisel and Bierwirth (2009a)
$cont\|dyn\|QCAP\|\sum res$	Meisel and Bierwirth (2006)
$cont\|dyn\|QCAP\|\sum wres$	Theofanis et al. (2007b)
$cont\|dyn\|QCAP,QCSP\|max(tard)$	Liu et al. (2006)
$cont\|dyn\|QCAP,QCSP\|max(compl),\sum(wait+hand)$	Meier and Schumann (2007)
$cont\|dyn\|QCAP,QCSP\|\sum(wait+hand+w_1tard)$	Ak and Erera (2006)
$cont\|dyn,due\|QCAP\|max(res)$	Hendriks et al. (2008)
$cont\|dyn,due\|QCAP\|\sum hand+w_1res$	Legato et al. (2008)
$cont\|dyn,due\|QCAP\|\sum(w_1wait+w_2res)$	Meisel and Bierwirth (2009b)
$hybr\|dyn\|fix\|\sum(w_1wait+w_2pos)$	Moorthy and Teo (2006) Dai et al. (2008)
$hybr\|dyn,due\|fix\|\sum wpos$	Chen and Hsieh (1999)
$hybr\|dyn\|pos\|\sum(wait+hand)$	Imai et al. (2007a)
$hybr\|dyn,due\|pos\|\sum w(wait+hand)$	Cordeau et al. (2005)
$hybr,draft\|dyn\|pos\|\sum(wait+hand)$	Nishimura et al. (2001)
$hybr,draft\|dyn\|pos\|max(compl),\sum wait,\sum order$	Cheong et al. (2007)
$hybr,draft\|dyn\|pos\|\sum pos$	Hoffarth and Voß (1994)
$hybr\|dyn\|pos,QCAP\|\sum wait,\sum tard$	Lokuge and Alahakoon (2007)

Lai and Shih (1992), Legato and Mazza (2001), Henesey et al. (2004), and Dragovic et al. (2005, 2006). Papers containing ideas and results published elsewhere, for example Kim (2005) and Crainic and Kim (2007), are also excluded.

As shown in Table 2, discrete and continuous problems are almost in balance, while dynamic problem formulations clearly prevail against static ones. The handling times of vessels are assumed to depend on the berthing positions in almost every discrete BAP formulation, because they are easily assessable in discrete models. However, only two continuous formulations and one half of the hybrid BAP formulations care for position based handling times. The QC resource is considered only in a few BAP formulations. Most models aim at the minimization of the average port stay time of vessels. Frequently addressed are also the minimization of tardy vessel departures and berthing positions different from desired berthing positions.

In the following, we briefly review the relevant papers according the grouping into discrete, continuous, and hybrid problems. Formulations with handling time characteristics *fix* or *pos* are merely summarized, whereas formulations that consider the crane resource within berth planning are presented in more detail.

3.2.1. Discrete problems

The discrete BAP has been studied in the static variant by Imai et al. (2001). In this problem the assignment and sequencing of vessels to berths is searched with respect to minimum waiting and handling times of the vessels. A Lagrangean relaxation based heuristic is presented to solve the problem. Hansen and Oğuz (2003) provide a more compact MIP formulation for the same problem. In the discrete static BAP considered by Imai et al. (1997) not only waiting and handling times of the vessels are minimized but also the deviation between the arrival order of vessels and the service order. The problem is reduced to a classical assignment problem, which is solved by the Hungarian method. In a recent paper of Imai et al. (2008b) the minimization of the weighted number of vessel rejections is handed over to a genetic algorithm (GA). A vessel is rejected at a terminal if it cannot be served without overshooting a due date, represented by a maximum acceptable waiting time.

Dynamic variants of the discrete BAP are considered by Imai et al. (2001), Hansen and Oğuz (2003), Monaco and Sammarra (2007), Imai et al. (2003), and Theofanis et al. (2007a) with regard to the port stay times of vessels. In the problem considered by Hansen et al. (2008) not only handling times but also service costs of vessels depend on the berth they are assigned to. Pursued is a tardiness objective which accounts for departure time related costs including penalties for tardiness as well as benefits for early departures. A Variable Neighborhood Search, used to solve the problem, turns out to be superior to the GA of Nishimura et al. (2001). Further departure time related objectives for the discrete dynamic BAP are proposed by Golias et al. (2006) and Imai et al. (2007b). In the model of Golias et al. (2007) arrival times and handling times of vessels are considered as stochastic variables. Since no specific performance measure is supposed for the problem, it is classified miscellaneous. A discrete dynamic BAP with due dates is formulated by Cordeau et al. (2005). A Tabu Search method is presented that outperforms the First-come First-served rule and also CPLEX. Mauri et al. (2008) design a column generation approach for the problem of Cordeau et al. (2005) which delivers better solutions in shorter runtime than Tabu Search. In the models of Han et al. (2006) and Zhou et al. (2006) the draft of vessels restricts the berth assignment decisions. In both papers, a GA is proposed to solve the problem. Moreover, Zhou et al. (2006) consider stochastic arrival and handling times of vessels and a waiting time restriction that is classified as a due date.

The crane resource is considered within a discrete BAP by Lee et al. (2006), Imai et al. (2008a), Liang et al. (2008), and Giallombardo et al. (2008). Lee et al. (2006) develop a GA to obtain berth plans that are evaluated by generating a feasible work plan for a given number of cranes at each berth. The evaluation bases on a modification of the QCSP model of Kim and Park (2004), which is not solved in the paper. Instead, a small-sized instance is provided as example. In the paper of Imai et al. (2008a) it is supposed that a

certain number of QCs has to be assigned to each vessel. The model of Liang et al. (2008) decides on the assignment of cranes to berths as well as on the berthing times and the positions of the vessels. In Giallombardo et al. (2008), cranes, berthing times, and berthing positions are assigned to vessels, such that the crane utilization (*res*) is maximized and the berthing position dependent container flow between pairs of vessels (*pos*) is minimized.

3.2.2. Continuous problems

The continuous static BAP with fixed vessel handling times has been introduced by Li et al. (1998). The problem is formulated as a "multiple-job-on-one-processor" scheduling problem. This allows to adapt the First-Fit-Decreasing heuristic, well-known from bin packing, for minimizing the maximum completion time among the vessels. Guan et al. (2002) propose to minimize the total weighted completion time of vessels for this type of problem and provide a priority rule based heuristic.

The continuous dynamic BAP with fixed handling times has been investigated in a number of studies. Guan and Cheung (2004) develop a tree search procedure to minimize the total weighted port stay time of vessels. In the problem of Wang and Lim (2007) the minimization of penalty cost for rejected vessels and apart berthing positions are the pursued objectives. A stochastic beam search algorithm is presented that is capable to solve instances with up to 400 vessels. A further objective, namely the minimization of tardiness of vessels, is treated by Moon (2000), Park and Kim (2002), Kim and Moon (2003), and Briano et al. (2005). Several solution methods are proposed for this problem, including a sub-gradient method (Park and Kim, 2002) and a Simulated Annealing approach (Kim and Moon, 2003). Lim (1998) formulates a problem, where the berthing times of vessels are already set by the arrival times. Instead, suitable berthing positions need to be determined and the goal is to minimize the maximum quay length required to serve vessels in accordance with the schedule. In this formulation of the problem, the goal is classified as a resource objective. The approach of Lim (1998) has been continued by Lim (1999), Tong et al. (1999), and Goh and Lim (2000).

The continuous BAP with handling times depending on berthing positions is studied by Imai et al. (2005) and Chang et al. (2008). Imai et al. (2005) suggest a heuristic solution method, which solves a discrete BAP first and then improves the obtained solution by shifting vessels along the quay as allowed in the continuous BAP. Chang et al. (2008) extend this model by incorporating the vessel draft in the berth allocation decision.

The continuous BAP in combination with crane assignment issues has been introduced by Park and Kim (2003). In this pioneering model it is simultaneously decided on berthing times, berthing positions, and the assignment of QCs to vessels at first. Here, the idea is to assign single QC-hours to the vessels such that variable-in-time QC-to-vessel assignments are possible. Next, it is decided on the specific cranes that serve a vessel. A Lagrangean relaxation based heuristic is used at the first decision level and dynamic programming is applied at the second level. A combined model for both decision levels is presented by Rashidi (2006).

The continuous static BAP with crane assignment issues is also studied by Oğuz et al. (2004). The pursued objective is to minimize the maximum completion time among the vessels. In contrast to Park and Kim (2003) and Oğuz et al. (2004) merely consider time-invariant QC-to-vessel assignments. However, crane productivity losses, which are frequently observed in practice, are anticipated in the made assignments. Unfortunately, the power of both approaches is not yet compared in the literature.

In the approach of Meisel and Bierwirth (2009a) both, variable-in-time QC-to-vessel assignments and crane productivity losses by interference, are modeled. Moreover, the crane productivity depends on the berthing positions of vessels. The objective is the minimization of service quality costs incurred by speedups of vessels and by tardy departures plus operational cost of the utilized QC-hours. To solve this complex model, squeaky wheel optimization and tabu search are proposed. Both deliver significant improvements against solutions reported by Park and Kim (2003) for a set of benchmark problems. In contrast to previous models, speeding up vessels is allowed only within certain bounds. For this reason an earliest possible arrival time is known for every vessel. Due to the dynamic arrival process of vessels, the problem is classified as dynamic.

A further approach aiming at the improvement of crane utilization is provided by Meisel and Bierwirth (2006). Here, a set of promising QC-to-vessel assignment patterns is generated for each vessel at first and then a priority rule is used to fix the berthing time, the berthing position, and the particular crane assignment pattern for every vessel. The minimization of cost for manning QCs is pursued. On this basis, several real-world berth plans are considerably improved without deteriorating the service quality substantially. Also Theofanis et al. (2007b) aim at an effective use of cranes by penalizing crane assignments that do not meet a targeted productivity rate. As a common solution method, a GA is proposed for the problem, which does not deliver satisfying results yet.

In a couple of papers, crane assignment and crane scheduling are involved to estimate vessel handling times for the berth planning. Liu et al. (2006) propose to derive vessel handling times from QCSP schedules which are generated in a preprocessing. Berthing times and QC-to-vessel assignments are subsequently determined for the vessels with respect to given berthing positions and the projected handling times. The minimization of the maximum relative tardiness of vessel departures is considered as objective. Meier and Schumann (2007) generate a berth plan using the approach of Guan and Cheung (2004). Next, QC schedules are build for all vessels basing on the QCSP model of Zhu and Lim (2006). The gained handling times are used to revise the prior berth plan iteratively, which is controlled by a multi-agent system (MAS). However, the conducted computations indicate that this coordination works not yet effective in order to minimize service objectives. A monolithic model for continuous dynamic berth allocation, crane assignment, and crane scheduling is presented by Ak and Erera (2006). The captured subproblems are considered on a highly aggregated level, which enables applying Tabu Search for the minimization of the port stay time and the avoidance of penalty cost for tardy departures of the vessels.

Further approaches include due dates of vessels into a dynamic BAP, which restricts the service of vessels to time windows. Hendriks et al. (2008) consider berth allocation at a tactical level, where service time windows, quay space, and crane capacity are reserved for a set of periodically arriving vessels. The objective is to minimize the maximum crane capacity reserved for a period. Legato et al. (2008) propose to take berthing positions, berthing times, and due dates from a berth plan that is generated using the model of Park and Kim (2003). The task remaining is to assign the available cranes to vessels such that the total handling time of vessels and the number of utilized cranes are minimized. A heuristic procedure is sketched for this problem and demonstrated at an example. Meisel and Bierwirth (2009b) assume that vessel due dates are explicitly given by liner schedules. In their model, vessels not served completely within the time window leave the port at the prescribed due date, irrespective of any unprocessed transshipments. This policy, known as cut-and-run, is typically applied in highly congested terminals. In order to minimize vessel waiting times and unprocessed transshipments, the solution methods presented in Meisel and Bierwirth (2009a) are adopted.

3.2.3. Hybrid problems

The hybrid BAP with fixed handling times is investigated by Moorthy and Teo (2006), Dai et al. (2008), and Chen and Hsieh (1999). In Moorthy and Teo (2006) berthing areas of vessels are determined at a tactical level. The goal is to achieve robust berth plans with respect to stochastic perturbations of vessel arrivals. To identify the impact of vessel delays, the service processes of vessels are considered as activities and represented in a precedence graph which is analyzed using the project evaluation review technique. Dai et al. (2008) transfer the gained results to the operational level, where precise positions are searched within the projected berthing areas. For this problem a Simulated Annealing algorithm is proposed which minimizes vessel waiting time and the berthing position dependent container flow between pairs of vessels. In an early paper of Chen and Hsieh (1999), a MIP formulation is given for a same problem which also incorporates vessel due dates.

Hybrid BAP formulations with position dependent vessel handling times are studied from different perspectives in several papers. Imai et al. (2007a) investigate a hybrid BAP for indented berths whereas Cordeau et al. (2005) derive a hybrid problem from a discrete BAP where the quay is dynamically repartitioned. Additionally, the models of Nishimura et al. (2001), Cheong et al. (2007), and Hoffarth and Voß (1994) incorporate the draft of vessels. Although crane assignment issues are approached in Hoffarth and Voß (1994), a heuristic is proposed that does not involve crane assignments in the berth allocation planning. Crane assignment decisions are explicitly included in a hybrid BAP by Lokuge and Alahakoon (2007) and accompanying papers, see Lokuge and Alahakoon (2004, 2005), and Lokuge et al. (2004). In this problem, multiple vessels can be served at one berth at a time. A set of cranes, shared by simultaneously served vessels, is assigned to each berth. A MAS is designed where all decisions to be made are distributed among specialized software agents. The system performance is compared with an existing terminal operating system (Jaya Container Terminal, Sri Lanka), achieving considerable reductions regarding vessel waiting times and tardiness.

4. Quay crane scheduling

4.1. Classification scheme

As for berth planning problems, there is no classification scheme existing for QC scheduling problems so far. The proposed scheme classifies problems according to four attributes. The *task attribute* concerns the definition of tasks that represent the workload of the considered vessel. The *crane attribute* describes the availability of QCs at the vessel and the consideration of the crane movement speed. The *interference attribute* addresses the spatial constraints that are defined in a problem. These attributes have been described in detail in Section 2.3. A fourth attribute defines the *performance measure* for evaluating solutions to a problem. Each of the four attributes can take different values. They are listed in Table 3.

Most of the performance measures for crane schedules aim at short vessel handling times to allow for earliest possible departures. Minimizing the completion time of tasks (*compl*) serves the purpose of a short vessel handling time. The makespan of a schedule, i.e. the maximum completion time among all tasks, is of particular interest, because it determines the departure time for a vessel. An effective utilization of cranes leads also to short vessel handling times. Related performance measures are the finishing time of cranes (*finish*), the utilization rate of cranes (*util*), the throughput of cranes (*through*), and the time spent for moving cranes to other quay positions (*move*).

Table 3
A classification scheme for QCSP formulations.

Value	Description
1. Task attribute	
Area	Tasks refer to *areas* of bays
Bay	Tasks refer to single *bays*
Stack	Tasks refer to container *stacks* within a bay
Group	Tasks refer to *groups* of containers
Container	Tasks refer to single *containers*
prmp	Preemption of tasks is allowed
prec	Precedence relations among tasks are given
2. Crane attribute	
ready	Individual *ready* times of QCs are given
TW	Cranes are available at the vessel within hard time windows
pos	Initial (and final) *pos*itions of QCs are prescribed
move	Travel time for crane *move*ment is respected
3. Interference attribute	
cross	Non-*cross*ing of QCs is respected
save	*Safe*ty margins between QCs are respected
4. Performance measure	
compl	*Compl*etion time of a task
finish	*Finish*ing time of a QC
util	*Util*ization rate of a QC
through	*Through*put of a QC
move	*Move*ment of a QC

Like in the classification of berth allocation problems, a $\sum()$ function indicates a total performance measurement of either tasks or QCs, and a $max()$ function indicates a worst performance measurement. Objective functions with weighted tasks or cranes are indicated by a weight w, whereas weights w_1, w_2 etc. address a multi-objective function. As an example for the classification scheme consider a problem where the tasks consist of all loading and unloading operations of containers in bays. Assume further that initial positions and a moving speed are given for the cranes and non-crossing as well as safety margins must be respected. If the problem is to minimize the completion of the latest finished task, it is referred to as $Bay|pos, move|cross, save|max(compl)$ in the classification.

4.2. Problem classification

Table 4 gives a comprehensive survey of QCSP formulations from the literature. The classification comprises optimization models and verbally introduced problem descriptions by a best fit. The survey shows that most approaches define tasks on the basis of single bays or container groups. Non-crossing constraints are involved in the majority of research, whereas safety margins have much less been considered. Crane attributes are neglected in diverse studies, i.e. there are no ready times or time windows and also no movement speed given for the cranes. Most of the published QCSP formulations are dealing with the makespan criterion. The following survey takes up the distinction of models by the task attribute. Crane scheduling problems for complete bays and for container groups are considered next. The remaining approaches are presented together in a third subsection.

4.2.1. Scheduling of complete bays

In the QCSP with complete bays we are searching for a crane schedule where each bay is exclusively served (usually without preemption) by one QC. For a realistic consideration of this problem, the incorporation of the non-crossing condition is more or less inevitable. Basically, non-crossing of cranes is quite easy to assure by generating schedules, where all QCs have an identical moving direction along the vessel which is not changed during the service. According to Bierwirth and Meisel (2009) such schedules are referred to as unidirectional schedules.

Table 4
Overview of QCSP formulations.

Problem classification	Reference
$Area\|-\|cross\|max(util)$	Winter (1999) Steenken et al. (2001)
$Area\|-\|cross, save\|-\sum w\ through$	Lim et al. (2002, 2004a)
$Bay\|-\|cross\|max(compl)$	Lim et al. (2004b,c, 2007) Zhu and Lim (2006) Lee et al. (2007, 2008a)
$Bay\|-\|cross\|\sum w\ compl$	Lee et al. (2008b)
$Bay\|move\|cross\|\sum move$	Ak and Erera (2006)
$Bay\|pos, move\|cross, save\|max(compl)$	Liu et al. (2006)
$Bay, prmp\|-\|-\|\sum w\ compl$	Daganzo (1989) Peterkofsky and Daganzo (1990)
$Bay, prmp\|pos, move\|cross, save\|max(compl)$	Liu et al. (2006)
$Stack, prec\|-\|-\|max(compl)$	Goodchild and Daganzo (2004) Zhang and Kim (2009)
$Group, prec\|ready, pos, move\|cross,$ $save\|w_1\ max(compl) + w_2\ \sum finish$	Kim and Park (2004) Moccia et al. (2006) Sammarra et al. (2007)
$Group, prec\|ready, pos, move\|cross, save\|$ $\sum w\ finish + \sum w\ tard$	Tavakkoli-Moghaddam et al. (2009)
$Group, prec\|ready, pos, move\|cross, save\|max(compl)$	Bierwirth and Meisel (2009)
$Group, prec\|move\|cross\|max(compl)$	Ng and Mak (2006)
$Group, prec\|TW\|cross, save\|max(compl)$	Jung et al. (2006)
$Group, prec\|TW, pos, move\|cross, save\|max(compl)$	Meisel (2009)
$Container, prec\|-\|-\|max(compl)$	Meisel and Wichmann (2008)

Under the premise of unidirectional crane schedules, some results on approximation algorithms are presented for the QCSP in Lim et al. (2004b). Moreover, a reformulation of the considered QCSP as a constraint programming model is provided in Lim et al. (2004c). The authors show that every unidirectional schedule can be obtained from a certain assignment of tasks to QCs. Hence, the problem can be solved by exploring the space of task-to-QC assignments for which a constraint propagation method and a Simulated Annealing algorithm are presented. Recently, Lim et al. (2007) have shown for the QCSP with complete bays that there is always an optimal schedule among the unidirectional ones. This seminal result demonstrates that searching the space of unidirectional schedules is not a heuristic reduction of the problem. An exact method for the unidirectional problem delivers the optimal solution even if the premise of unidirectional crane schedules is dropped.

Liu et al. (2006) propose a MIP model for the QCSP which includes initial crane positions, moving speed, and interference conditions for the cranes. The structure of unidirectional schedules is anchored in this model, which allows to formulate the non-crossing condition and safety margins in a straight-forward manner. Furthermore, with the focus put on unidirectional schedules, the search space is significantly reduced which allows to solve non-trivial instances by a standard solver.

In a couple of papers, schedule unidirectionality is neither assumed in models nor in algorithms. A Branch-and-Bound method with limited capability and a better performing Simulated Annealing algorithm are presented for the QCSP with complete bays by Zhu and Lim (2006). A GA and a greedy algorithm are developed by Lee et al. (2007, 2008a) for the same problem. The approach is augmented in Lee et al. (2008b) by replacing the makespan criterion with the total weighted completion time of tasks. Ak and Erera (2006) treat the QCSP for a set of vessels that are served simultaneously at the quay. The assignment problem of cranes to the bays of the vessels is modeled as a min-cost flow problem, with the QC travel time between vessels to be minimized.

Scheduling of QCs with preemption allowed has been investigated by Daganzo (1989). The idea is to assign cranes to bays for certain time slots, such that the overall workload is well balanced for the cranes. As a consequence, a bay might be served consecutively by different QCs. Note that this early work does not take crane interference into consideration. The goal is to minimize the total weighted completion times of vessels. The considered scheduling problem is solved by rules of thumb and by a Branch-and-Bound method later proposed by Peterkofsky and Daganzo (1990). Preemptive schedules are also allowed in another version of the above mentioned model of Liu et al. (2006). The authors use this model to prove by experiment that the sharing of bays among cranes can significantly improve the makespan of a QCSP instance.

4.2.2. Scheduling of container groups

Enabling the cranes to share the workload of bays is the most typical feature of the QCSP with container groups. For this reason the need to avoid crane crossing is even more stressed in this problem than in the QCSP with complete bays. The QCSP with tasks defined on the basis of container groups has been introduced by Kim and Park (2004). Their model considers QC operations in detail by taking precedence relations among tasks, crane attributes, and crane interference into account. In this model, safety margins between QCs are enforced by a non-simultaneity constraint between tasks located in adjacent bays. The pursued objective is the minimization of the weighted sum of makespan and QC finishing times. The authors propose a Branch-and-Bound method and a Greedy Randomized Adaptive Search Procedure (GRASP). The Branch-and-Bound method outperforms GRASP in terms of solution quality but fails for larger test problems.

The model of Kim and Park (2004) has been refined by Moccia et al. (2006), leading to a more stringent problem formulation. They develop a Branch-and-Cut algorithm which significantly improves solutions for the benchmark suite provided in Kim and Park (2004). Sammarra et al. (2007) present a Tabu Search algorithm for the same problem where a neighborhood is defined by resequencing the tasks of a crane and by swapping tasks between cranes. Compared to the Branch-and-Cut of Moccia et al. (2006), Tabu Search cuts down the computation time significantly for the larger instances of the benchmark suite at the expense of a slightly weaker solution quality. Tavakkoli-Moghaddam et al. (2009) extend the model of Kim and Park (2004) towards scheduling cranes for a set of vessels in parallel. A GA is presented for the minimization of the total weighted finishing time of QCs and the total weighted tardiness of vessels (as known from berth planning). Unfortunately, the GA's performance is not compared with other algorithms proposed in this stream.

In Bierwirth and Meisel (2009) a Branch-and-Bound algorithm is presented for searching a reduced solution space of the problem. The idea is to solely rely on unidirectional schedules. Different to the QCSP with complete bays, the authors show that there is not necessarily a unidirectional schedule among the optimal schedules to a problem instance. Hence, the proposed reduction makes the used Branch-and-Bound algorithm actually a heuristic. Still, the algorithm clearly outperforms the heuristics of Kim and Park (2004), Moccia et al. (2006), and Sammarra et al. (2007) in terms of solutions quality and computation time.

Ng and Mak (2006) obtain a QCSP with container groups by including the import containers and the export containers of a bay into two separate groups. A precedence relation is inserted between the two tasks of each bay. The problem is solved by partitioning the set of bays into areas such that sharing of workload by cranes is only possible at the border of two areas.

Using time windows for the service of cranes is discussed by Jung et al. (2006) but not outlined in detail. A mathematical

problem formulation that includes time windows in the QCSP with container groups is presented by Meisel (2009). In this model QCs can possess multiple time windows for serving a vessel, as observed in practice under a variable-in-time QC-to-vessel assignment. The Branch-and-Bound algorithm of Bierwirth and Meisel (2009) is adapted to this problem for solving crane scheduling in combination with berth allocation and crane assignment.

4.2.3. Other problems

Taking a look at Table 4, it can be seen that defining tasks on the basis of bay areas, container stacks, or single containers is rather seldom dealt with in the scientific literature. Winter (1999) and Steenken et al. (2001) assign bay areas to QCs such that the maximum difference regarding the utilization of any two cranes is minimized. The authors show that crane scheduling on the basis of bay areas leads to a partitioning problem that can easily be solved optimal for instances of practical size. Lim et al. (2002, 2004a) assign bay areas to QCs assuming individual throughput rates for the cranes. They aim at the maximization of the total throughput and propose several heuristics, where a squeaky wheel optimization method combined with a local search performs best. Note that none of the mentioned approaches takes detailed crane schedules into consideration.

A stack-based QCSP model has been introduced by Goodchild and Daganzo (2004) and is further studied in a number of papers by Goodchild and Daganzo (2005a,b, 2006, 2007) and Goodchild (2006). The basic idea is to consider one crane processing the container stacks of one bay. Two precedence-related tasks are defined for each container stack, one for unloading the stack and one for loading the stack. Note that the loading and unloading of different stacks can be parallelized in the crane schedule, which is referred to as double cycling. The problem is to find a sequence for processing the stacks, which minimizes the makespan. The authors reformulate this problem as a two-machine flow shop scheduling problem which is solved to optimality using the rule of Johnson (1954). The approach is continued by Zhang and Kim (2009) who modify Johnson's rule in order to handle hatches that cover adjacent stacks. Double cycling is also addressed by Meisel and Wichmann (2008), who deal with crane scheduling on the basis of single containers. Contrasting the approach of Goodchild and Daganzo, reshuffle containers can be repositioned in the bay instead of temporarily unloading them, which accelerates the service process. A GRASP heuristic is used to solve the resulting scheduling problem.

QC scheduling is also considered within other operations planning problems concentrating e.g. on stowage planning, horizontal transport operations, and yard crane scheduling, cf. Gambardella et al. (2001), Bish (2003), Kim et al. (2004), Lee et al. (2005), Imai et al. (2006), Chen et al. (2007), and Canonaco et al. (2008). Since the primary focus of this research is not on the determination of crane schedules, we do not explicitly address these approaches in the classification.

5. Integrated seaside operations planning

5.1. Classification scheme

Recent integration approaches for seaside operations planning motivate a further classification scheme, based on the concepts briefly introduced in Section 2.4. To distinguish problem integration by monolithic models (deep integration), by problem preprocessing, and by feedback loops, the notation of Table 5 is used. In this table, capitals A and B stand proxy for a BAP, QCAP, or QCSP. If a planning problem involves multiple decision variables but not all of them are determined at once in the integration model, the addressed decisions appear in brackets. For example, $\boxed{\text{BAP, QCAP(number)}} \rightarrow$ QCAP (specific) stands for an integration where a berthing time and position, and a number of cranes are assigned to every vessel in a monolithic model, while the used cranes are specified subsequently.

5.2. Problem classification

Table 6 gives an overview of approaches for integrated seaside operations planning. In the following we review the mentioned papers with respect to the used integration models. Approaches that exclusively use functional integration are proposed by Lee et al. (2006) and Lokuge and Alahakoon (2007). Lee et al. (2006) consider a feedback loop integration between the discrete BAP and the QCSP. There is no QCAP involved because the berths possess dedicated cranes. The solution of the BAP delivers a sequence for serving the vessels at each berth. A resulting QCSP is solved for every vessel and the obtained handling times are returned to the berth planning level to revise the vessel sequences. This loop is executed for a preset number of iterations. In Lokuge and Alahakoon (2007) a MAS is used to integrate the hybrid BAP and the QCAP. Cranes are dedicated to berths which are shared by vessels served at the same time. The problem is solved by software agents responsible for berth planning and communicating with other agents responsible for the crane assignment. The architecture of the used MAS constitutes a feedback loop integration of the BAP and the QCAP.

A deep integration of the continuous BAP and the QCAP is investigated by Oğuz et al. (2004), Meisel and Bierwirth (2006, 2009a,b), Hendriks et al. (2008), Park and Kim (2003), Rashidi (2006), and Theofanis et al. (2007b). These papers present optimization models to decide on the berthing time, the berthing position, and the number of cranes for each vessel. The same decisions are considered for the discrete BAP by Giallombardo et al. (2008), Liang et al. (2008), and Imai et al. (2008a). In four of these approaches, the specific cranes used for the service of vessels are additionally determined.

Table 5
A notation scheme for problem integration concepts.

Notation	Description
$\boxed{\text{A, B}}$	Deep integration of problems A and B
A → B	A is preprocessed to B
A ⇄ B	Feedback loop integration of A and B

Table 6
Overview of integration concepts for seaside operations planning.

Integration concept	Reference
BAP ⇄ QCSP	Lee et al. (2006)
BAP ⇄ QCAP	Lokuge and Alahakoon (2007)
$\boxed{\text{BA P, QCAP(number)}}$	Oğuz et al. (2004) Meisel and Bierwirth (2006, 2009a,b) Hendriks et al. (2008) Giallombardo et al. (2008) Liang et al. (2008)
$\boxed{\text{BAP, QCAP(number)}} \rightarrow$ QCAP(specific)	Park and Kim (2003)
$\boxed{\text{BAP, QCAP(number)}} \rightleftarrows$ QCAP(specific)	Imai et al. (2008a)
$\boxed{\text{BAP, QCAP}}$	Rashidi (2006) Theofanis et al. (2007b)
BAP ⇄ $\boxed{\text{QCSP, QCAP}}$	Meier and Schumann (2007)
$\boxed{\text{BAP, QCAP(number), QCSP}} \rightarrow$ QCAP (specific)	Ak and Erera (2006)
QCSP → $\boxed{\text{BAP(berthing times), QCAP}}$	Liu et al. (2006)
QCSP → $\boxed{\text{BAP, QCAP}}$ ⇄ QCSP	Meisel (2009)
$\boxed{\text{QCAP, QCSP}}$	Daganzo (1989) Peterkofsky and Daganzo (1990) Tavakkoli-Moghaddam et al. (2009)

For this purpose different integration concepts are applied. Park and Kim (2003) consider the specific crane assignment as an end-of-pipe optimization which is appropriately solved in a post-processing phase. In contrast, Imai et al. (2008a) return the specific crane assignment to the berth planning level where the made berthing decisions are evaluated and possibly revised. In two papers, the number and the specific set of cranes assigned to vessels are decided within a monolithic model. Rashidi (2006) merges the top-level problem and the end-of-pipe problem of Park and Kim (2003) into a single optimization model. A deep integration is also proposed by Theofanis et al. (2007b) for simultaneously assigning QCs and allocating vessels along the quay.

Several authors study integration models that involve all of the three seaside planning problems. A feedback loop integration of the BAP and the QCSP is described by Meier and Schumann (2007). The loop propagates a berth plan to the crane scheduling level. Detailed vessel handling times are obtained and returned to the top-level for an adjustment of the berth plan. The approach comprises a deep integration of QCAP and QCSP, because the crane schedules are collectively built for vessels served at the same time. Ak and Erera (2006) present an integration model that jointly decides on berth allocation, crane assignment, and crane scheduling. Merely the specific crane assignment is determined in a postprocessing, as has been proposed by Park and Kim (2003). The integration model of Liu et al. (2006) targets on the revision of a tentative berth plan. First, crane schedules are preprocessed to generate possible vessel handling times for each vessel and each assignable number of cranes. Next, specific cranes are assigned to vessels, where the tentative handling times are replaced by selecting values provided in the preprocessing phase. In order to minimize the maximum vessel tardiness, the tentative berthing times are revised in this model. The berthing positions are taken from the tentative berth plan. A three-phase integration model has been developed by Meisel (2009). Similar to Liu et al. (2006), QCSP instances are solved in a preprocessing for each vessel and each assignable number of cranes. In contrast, Meisel (2009) gathers the crane productivity rates instead of handling time data. In the central optimization phase, berth planning and crane assignment are conducted with respect to possible crane productivity rates by using the algorithms of Meisel and Bierwirth (2009a). Finally, a feedback loop is needed for the adjustment of the berth plan, the crane assignment, and the crane schedules. Computational tests show that this approach clearly dominates the traditional sequential planning process which is depicted in Fig. 5.

Further integration models are formulated by Daganzo (1989) and Peterkofsky and Daganzo (1990). They combine the QCAP and the QCSP by simultaneously scheduling multiple cranes for a set of vessels. The authors remark that berthing decisions should be integrated with crane operations planning and illustrate this issue by examples under the assumption of identical sized vessels. Also Tavakkoli-Moghaddam et al. (2009) deal with the integration of crane assignment and scheduling. In this work the QCSP model of Kim and Park (2004) is extended such that multiple vessels are considered in parallel.

6. Summary

In this survey we have reviewed all research streams which are known to us in the field of seaside operations planning. The provided classification of existing models shows that the determination and the consideration of vessel handling times is the most crucial yet difficult need for a proper planning of seaside operations. The impact of berthing positions on the handling times is well-established in discrete and hybrid BAP formulations, but hardly considered in the continuous case so far. Especially the interdependencies between the QC resource utilization and the vessel handling times receive much attention in current research. Detailed QCSP models incorporating e.g. crane interference effects are developed to obtain reliable estimates about port stay times and the transshipment capacity of a terminal. All this explains the growing interest in integration models that allow finding well balanced decisions for the involved subproblems. We hope that the provided problem classification helps readers facing a particular problem to identify suitable modeling techniques and efficient optimization methods.

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